Gurdarshan Singh

Patiala, Punjab | Email | LinkedIn | GitHub

Summary

Computer Engineering undergraduate with strong foundations in Data Structures & Algorithms, Object-Oriented Programming, Concurrency, and Software Engineering. Skilled in Java, Python, and JavaScript, with hands-on experience designing and implementing high-performance backend systems, including real-time collaboration platforms, multi-agent orchestration engines, and mathematical optimization solvers. Proven ability to build scalable, maintainable software across the full development lifecycle using Flask, REST APIs, SQL, and Git.

Education

Thapar Institute of Engineering and Technology — Patiala, Punjab
B.Tech. in Computer Engineering | CGPA: 8.00/10 — *Aug. 2023 – May 2027*

Pepsu International Public School — Patiala, Punjab
Higher Secondary Education (Science – PCM) — *April 2021 – March 2023*

Experience

Software Engineering Research Intern | ELC (Experiential Learning Centre) — Patiala, Punjab
Thapar Institute of Engineering and Technology — *May 2024 – July 2024*

- Designed and developed a Flask-based backend application integrating facial expression recognition and speech signal processing into a unified software platform.
- Built modular software components and optimized data processing pipelines, improving stress classification accuracy by 12% while ensuring maintainable and scalable code.
- Collaborated with faculty mentors throughout the software development lifecycle, including system design, implementation, testing, deployment, and technical presentations.

Projects

SyncBoard | *JavaScript, SQLite, WebSockets, CRDTs* — 2026
- Built a framework-less, real-time collaborative whiteboard and code editor using a custom RGA CRDT for conflict-free sync and an event-sourced SQLite backend enabling deterministic session replay.

Fleetwise | *Java, Multithreading, Graph Algorithms* — 2026
- Developed a zero-dependency Java multi-agent orchestration engine for warehouse robot fleets, combining Conflict-Based Search and the Hungarian algorithm for collision-free, globally optimal concurrent routing.

Aegis | *Java, Concurrency, JVM Internals* — 2025
- Built a zero-dependency JVM resilience gateway implementing lock-free rate-limiting algorithms and an atomic circuit breaker using core Java concurrency primitives (CAS loops, Atomic variables).

PlaceCard | *JavaScript, Node.js, Optimization Algorithms* — 2025
- Designed a vanilla JS and Node.js seating-planner that models complex logistical constraints as an NP-hard graph-partitioning problem, solved using Simulated Annealing and Union-Find clustering.

Perk Arbitrage Optimizer | *Python, Linear Programming, Serverless* — 2025
- Built a serverless Python engine applying Binary Integer Linear Programming (BILP) via the CBC solver to mathematically maximize credit card reward returns under real-world spending constraints.

Technical Skills

Programming Languages: Java, Python, JavaScript, C/C++, SQL
Core Computer Science: Data Structures & Algorithms, Concurrency & Multithreading, Graph Algorithms, Optimization (Linear Programming, Simulated Annealing), Object-Oriented Programming, Database Management Systems, Software Engineering
Backend & Systems: Flask, Node.js, REST APIs, SQL, SQLite, Event-Sourced Architecture, CRDTs, WebSockets
Developer Tools: Git, GitHub, Docker, VS Code, Jupyter Notebook
Frameworks & Libraries: Flask, Streamlit, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas

Certifications

Build a CI/CD Pipeline with Docker: From Code to Deployment – Coursera
NVIDIA: Fundamentals of Deep Learning – NVIDIA
Semantic Segmentation with Amazon SageMaker – Coursera